Numerical Valuation of Perpetual American Put Options via LCP

An M-matrix discretisation solved by policy iteration, with every number tied to a residual or an oracle

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Outline

1. Model
2. Method
3. Verification
4. Extensions
5. Conclusions

The Problem: an Unknown Exercise Region

- American put: sell at K at any time; the value is an optimal-stopping value.
- Perpetual limit: $\partial_t V = 0$, so the pricing problem becomes an elliptic obstacle problem on $[0, S_{\max}]$.
- Exercise set $[0, S^*]$, continuation region (S^*, ∞); V and V' are continuous at S^*.
- The boundary S^* is part of the answer, not an input.

Takeaway

The difficulty is the unknown free boundary S^*, not the Black–Scholes operator.

Model, Boundary Data, and the Benchmark

$$\mathcal{L}V = \tfrac{1}{2}\sigma^2 S^2 V'' + (r-q)SV' - rV$$

$$\min\{-\mathcal{L}V,\ V-\psi\} = 0, \qquad \psi = (K-S)^+, \qquad V(0) = K,\ \ V(S_{\max}) = 0$$

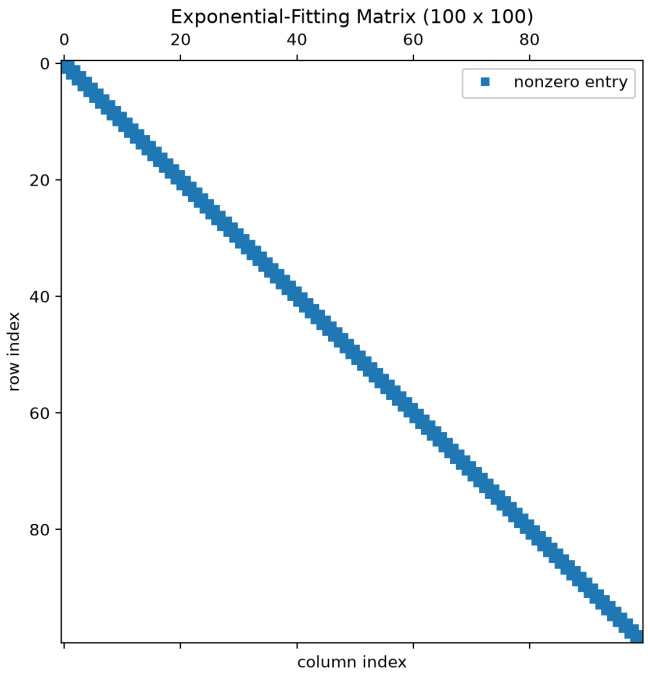
- **Stage 1:** $K = 100,\ r = 0.05,\ q = 0,\ \sigma = 0.20,\ S_{\max} = 2400.$
- **Stage 1 benchmark:** $\lambda_- = -2.5,\ S^* = 71.4285714286,$ $V(S) = (K - S^*)(S/S^*)^{\lambda_-},\ V(100) = 12.3200328678.$
- **Stage 2:** same $K, r, S_{\max},\ q = 0.02,$ $\sigma(S) = 0.20\,[1 + 0.35\tanh(\log(S/K)/0.40)].$
- **2D spread:** same $K, r,\ q_1 = 0,\ q_2 = 0.02,\ \sigma_1 = \sigma_2 = 0.20,\ \rho = 0,$ $L = 300.$

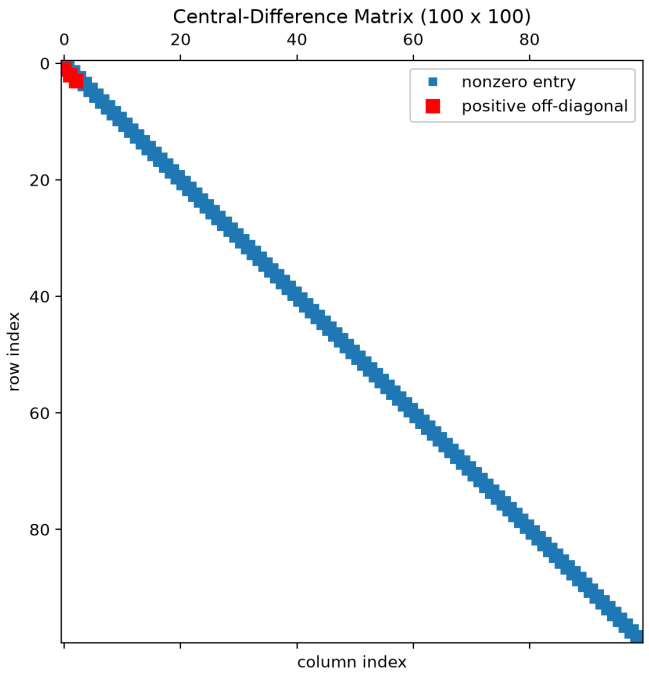
Takeaway

Constant coefficients give a closed-form oracle, so Stage 1 is judged against an answer, not against a prettier plot.

Central Differencing Destroys Monotonicity

$$\mathrm{Pe}_i = \frac{b_i h}{a_i}, \qquad \mathrm{Pe}_1 = \frac{2(r-q)}{\sigma^2} = 2.5 > 2 \quad \text{(independent of } h\text{)}$$





Takeaway

A positive off-diagonal breaks the M-matrix; refinement never restores it.

Exponential Fitting Restores the M-Matrix

$$B(z) = \frac{z}{e^z - 1} > 0, \qquad \ell_i = -\frac{a_i}{h^2} B(\mathrm{Pe}_i), \quad u_i = -\frac{a_i}{h^2} B(-\mathrm{Pe}_i)$$

$$d_i = \frac{a_i}{h^2}\left[B(\mathrm{Pe}_i) + B(-\mathrm{Pe}_i)\right] + r, \qquad d_i + \ell_i + u_i = r > 0$$

- Off-diagonals are negative and the diagonal is positive at every Péclet number.
- Row sum is exactly r, so the tridiagonal matrix is a nonsingular M-matrix.
- No limiter and no start-up correction; at $\mathrm{Pe} = 0$ the weights reduce to central differences.

Takeaway

One Bernoulli weight makes the operator monotone for every mesh and every Péclet number.

Discrete LCP and Policy Iteration

$$\min\{(A_h V)_i - F_i,\ V_i - \psi_i\} = 0, \qquad i = 1, \ldots, N-1$$

- Start from $V = \psi$; a row is either active (obstacle) or free (PDE).
- Policy step: put $V_i = \psi_i$ on active rows, zero their couplings, and solve the sparse tridiagonal system for the free values.
- Release active rows with $a_{h,i} < 0$; project $V \leftarrow \max(V, \psi)$.
- Stop when the active set is unchanged and $\max_i |\Delta V_i| < 10^{-12}$.
- Diagnostics on every mesh: $\min a_h$, $\min b_h$, $\| \min(a_h, b_h)\|_\infty$, $\psi \le V \le K$, monotone, convex, one contact set.

Takeaway

Policy iteration turns the nonlinear complementarity problem into a sequence of exact linear solves; the active set is the only unknown.

Verification Protocol

- **Oracle:** closed-form S^* and $V(S)$ for constant coefficients.
- **Residual:** $a_h = A_h V - F$, $b_h = V - \psi$, and $\| \min(a_h, b_h) \|_\infty$.
- **Shape:** $\psi \le V \le K$, monotone decrease, convexity, one contiguous contact set.
- **Second solver:** PSOR on the identical operator, tolerance, and initial iterate.
- **Aligned grid:** a node at S^* separates stencil order from the phase of the kink.

Takeaway

Every numerical claim in the report is tied to a residual, an oracle, or a second method.

Stage 1: the Fitted Stencil is Second Order

Node at S^*, piecewise uniform grid, asymptotic Robin closure.

N	h	$E_N = \max_i \lvert V_i - V(S_i)\rvert$	error ratio	observed p
600	4.0	1.0732×10^{-2}	—	—
1200	2.0	2.6895×10^{-3}	3.990	1.997
2400	1.0	6.7222×10^{-4}	4.001	2.000
4800	0.5	1.6818×10^{-4}	3.997	1.999

- LCP residual $\leq 5.5 \times 10^{-12}$ on every mesh; policy update $< 10^{-12}$.

Takeaway

Halving h divides the error by almost exactly four: the observed order is 2.00.

Stage 1: Production Contact Brackets

Uniform grid, production condition $V(S_{\max}) = 0$.

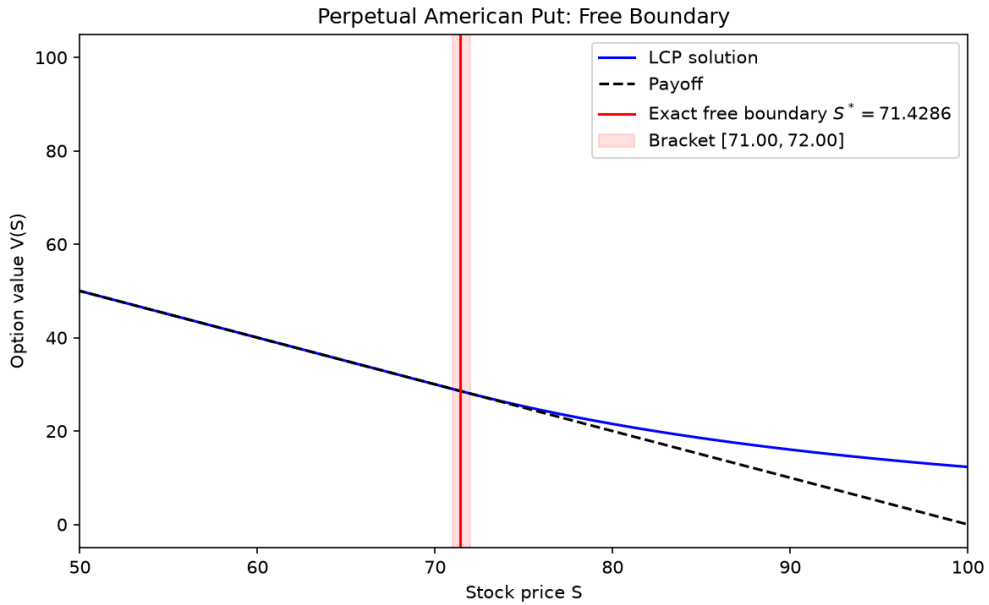
N	h	contact bracket	S_h^*	$\lvert S_h^* - S^* \rvert$	E_N
600	4.0	$[72.0, 76.0]$	72.000	5.714×10^{-1}	7.905×10^{-3}
1200	2.0	$[72.0, 74.0]$	72.000	5.714×10^{-1}	7.905×10^{-3}
2400	1.0	$[71.0, 72.0]$	71.654	2.254×10^{-1}	4.366×10^{-3}
4800	0.5	$[71.5, 72.0]$	71.500	7.143×10^{-2}	4.366×10^{-3}

- The bracket is the transition from the last active node to the first continuation node.
- At $N = 600, 1200, 4800$ it lies to the right of S^*; only $N = 2400$ contains it.

Takeaway

A contact bracket is a discrete transition, not a confidence interval; the coarse ones miss S^* and are reported as such.

Stage 1: Free Boundary at $N = 2400$

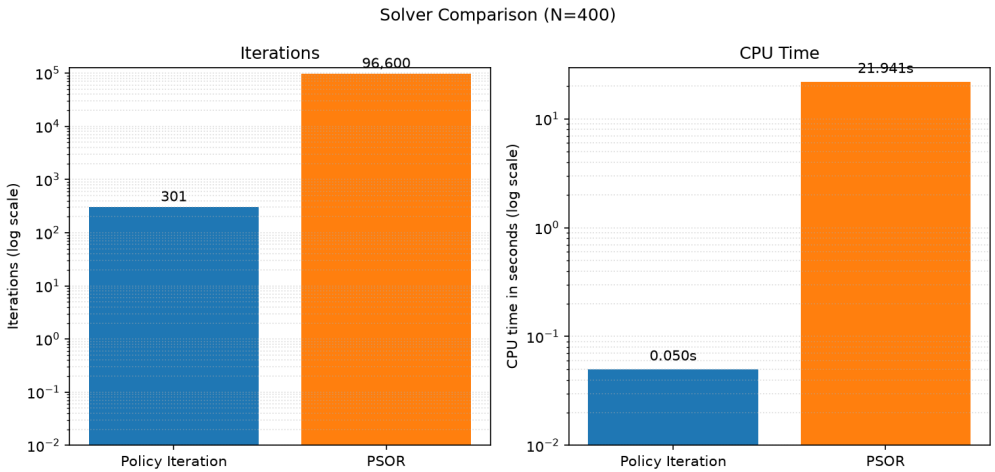


- Bracket $[71, 72]$ contains $S^* = 71.4286$ (estimate 71.654).
- The continuation branch leaves the payoff with slope -1.

Takeaway

The boundary comes from the obstacle slack, not from an imposed condition.

Algebraic Error is Separated from Discretisation Error



- 301 sweeps, 0.045 s vs PSOR $96{,}600$ sweeps, 20.635 s; LCP residuals 9.1×10^{-13} vs 9.9×10^{-11}; solutions differ by 5.7×10^{-10}.

Takeaway

The algebraic difference is far below the discretisation error, so the reported error is spatial.

Stage 2: Local Volatility vs. an ODE Reference

$\sigma(S) = 0.20\,[1 + 0.35\tanh(\log(S/K)/0.40)]$, $q = 0.02$, $S_{\max} = 2400$.

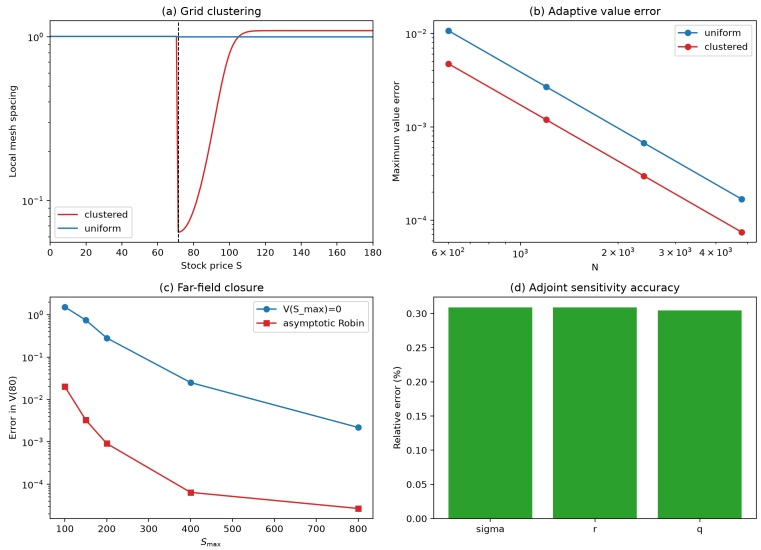
- DOP853 shooting: $S^* = 69.1664$ (infinite), 69.1702 (finite), $V(100) = 16.2055$; production $S_h^*(2400) = 69.1688$.

N	aligned E_N	ratio	observed p	production bracket
600	3.2218×10^{-2}	—	—	$[68, 72]$
1200	8.0948×10^{-3}	3.980	1.993	$[70, 72]$
2400	2.0239×10^{-3}	4.000	2.000	$[69, 70]$
4800	5.0610×10^{-4}	3.999	2.000	$[69, 69.5]$

Takeaway

The local-vol solution is checked by an ODE reference that shares no code with the LCP, and the order is still two.

Adaptive Grid and Far-Field Closure



- Clustering: $2.255\times$ smaller max error at $N = 2400$; clustered ratios $3.982, 4.002, 3.999$.
- Robin at $S_{\max} = 400$: $V(80)$ error $2.507 \times 10^{-2} \to 6.423 \times 10^{-5}$.

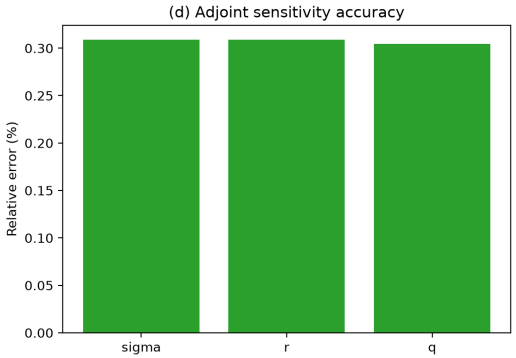
Takeaway

Clustering lowers the constant; Robin removes the far-field bias; the solver is unchanged.

Extensions: Adjoint and 2D Spread Put

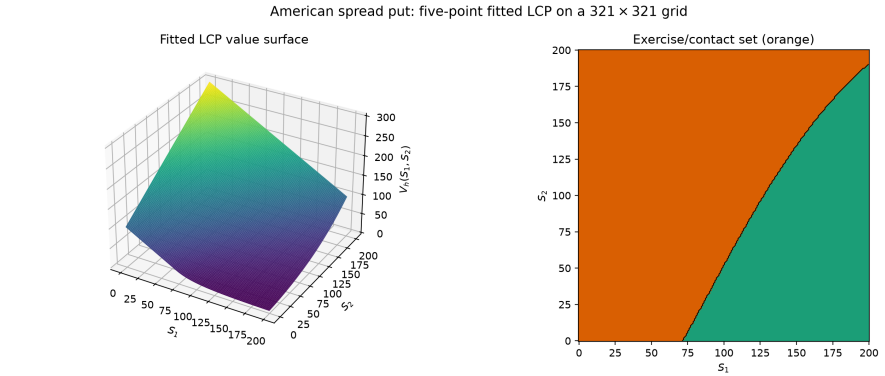
Adjoint sensitivities

- One transposed solve gives $\partial V(100)/\partial\theta$ for $\theta \in \{\sigma, r, q\}$.
- Matches direct finite differences to 2×10^{-9}; 0.31% from the exact derivative.



American spread put ($\rho = 0$)

- Fitted five-point operator stays an M-matrix.
- $101{,}761$ unknowns; error $4.07 \times 10^{-2} \to 1.61 \times 10^{-2}$; $L = 300$ vs 400: 9.7×10^{-4}.



Takeaway

The same fitted construction provides parameter risk at one extra solve and scales to a genuine two-dimensional obstacle.

Limitations: What We Do Not Claim

- **Far field:** $V(S_{\max}) = 0$ is a truncation; the pure truncation in $V(100)$ is 5.578×10^{-3}, while the total error is 4.055×10^{-3} after partial cancellation.
- **Kink:** contact-set localisation is at most first order; coarse brackets can miss S^*.
- **Stage 2 reference:** semi-analytic (ODE shooting), not a closed form.
- **2D:** only $\rho = 0$; a correlated model needs a nine-point stencil.

Takeaway

Verified, approximated, and not claimed are kept separate: the truncation and localisation errors are measured, not hidden.

Conclusions

- Exponential fitting is an M-matrix at every cell Péclet number; no limiter is needed.
- Policy iteration plus sparse direct solves: 301 vs $96{,}600$ sweeps at $N = 400$.
- Stage 1 order 2.00 (ratios 3.99–4.00); Stage 2 order 2.00 against an independent ODE reference.
- Algebraic error $\leq 10^{-12}$ against discretisation errors of 10^{-4}–10^{-2}.
- Extensions: clustering ($2.25\times$), Robin closure ($390\times$), adjoint Greeks, 2D spread LCP.

Thank You

- Questions welcome.
- Every figure and table regenerates with `python run_all.py`.